

Lesson 9: Sequences

Assessment Rubric for Infinite Sequences

Criteria	4 - Mastery	3 - Proficiency	2 - Approaching	1 - Beginning
Interpretation & Representation	Accurately identifies terms, general nth term (a_n), and recursive formulas.	Identifies the general term with minor notational errors.	Struggles to define the nth term from a list of numbers.	Cannot identify the pattern or term formula.
Convergence/ Divergence	Correctly determines if the sequence converges or diverges.	Determines convergence but with minor justifying errors.	Struggles to distinguish between sequence and series convergence.	Cannot determine if the sequence converges.
Limit Evaluation	Correctly calculates $\lim_{n \rightarrow \infty} a_n$ using limit laws or L'Hôpital's rule.	Calculates limits correctly, with minor algebraic errors.	Attempts limits but struggles with ∞/∞ forms.	Unable to evaluate limits.

Behavioral Analysis	Confidently proves boundedness and monotonicity (increasing/decreasing).	Correctly identifies behavior but provides incomplete proofs.	Identifies behavior incorrectly or inconsistently.	Cannot determine if the sequence is bounded or monotonic.
Justification	Thoroughly justifies convergence using appropriate theorems (e.g., Squeeze Theorem).	Uses correct tests but justification is lacking in detail.	Applies incorrect convergence tests.	Provides no justification for findings.
